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**Exercice 1 – Boosting**


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Reminder of the algorithm : we aim to construct a combination of weak classifiers  $f_T(x) = \sum_{t=1}^T \alpha_t h_t(x)$  iteratively, in such a way as to better account for the errors of previous iterations at each step. To do this, a weight distribution over the examples is considered and adjusted at each iteration to increase the weight of misclassified examples and decrease the weight of well-classified examples. Let  $D_t = (w_t(1), \dots, w_t(n))$  be the distribution of the example weights at step  $t$ , with  $D_1$  corresponding to the uniform distribution. The algorithm consists of iterating the following procedure :

1. Choose  $h_t$  that minimizes the error according to  $D_t$
2. Compute the error  $\epsilon_t$  associated with the classifier  $h_t$  according to  $D_t$
3. Set  $\alpha_t = \frac{1}{2} \ln(\frac{1-\epsilon_t}{\epsilon_t})$
4. Update  $D_{t+1} : w_{t+1}(i) = \frac{1}{Z_t} w_t(i) e^{(-\alpha_t y_i h_t(x_i))}$ , where  $Z_t$  is the normalization factor

**Q 1.1** Recall the principle and the differences between boosting and bagging. Let the following dataset be given :  $Y^+ = \{(-3, -1), (-3, 1), (3, -1), (3, 1)\}$ ,  $Y^- = \{(-1, -1), (-1, 1), (1, -1), (1, 1)\}$ . Considering stumps (functions of the form  $\mathbb{1}_{x_i < \theta_i}$ , corresponding to a decision tree with 2 leaves) as weak classifiers, what are the first two classifiers learned ? Are they sufficient for perfect classification ?

**Q 1.2** Express the error  $\epsilon_t$  in terms of a given cost function  $l(x, y)$  and the  $w_t(i)$ .

**Q 1.3** How does  $\alpha_t$  vary in terms of  $\epsilon_t$  ? What happens to  $w_{t+1}(i)$  if example  $i$  is well-classified ? Misclassified ?

We will show below that the algorithm indeed optimizes the training error. The demonstration principle consists of showing that at each step  $t$ , the error is bounded by  $Z = \prod_{j=1}^t Z_j$ , and this product converges to 0.

**Q 1.4** We will now show that the choice of  $\alpha_t$  leads to minimizing  $Z_t$ .

**Q 1.4.1** Express  $Z_t$  and  $\epsilon_t$  in terms of  $w_t(i)$ ,  $\alpha_t$ , and  $y_i h_t(x_i)$ .

**Q 1.4.2** Express  $\frac{\partial Z_t}{\partial \alpha_t}$ . Deduce the value of  $\alpha_t$  that minimizes  $Z_t$ .

**Q 1.4.3** Give the expression of  $Z_t$  in terms of  $\epsilon_t$  for optimal  $\alpha_t$ .

**Q 1.4.4** Let  $\gamma_t = \frac{1}{2} - \epsilon_t$ . Knowing that  $1 - x \leq e^{-x}$ , show that  $Z$  decreases exponentially as a function of  $t$ .

**Q 1.5** Now we will show that  $Z$  is an upper bound for the 0-1 error.

**Q 1.5.1** Express  $w_{t+1}(i)$  in terms of  $h_j(x)$ ,  $\alpha_j(i)$ ,  $Z_j$ ,  $1 \leq j \leq t$ , and then in terms of  $f_t(x_i)$ . Deduce an expression for  $\sum_i w_t(i)$  in terms of  $Z_j$  and  $y_i f_t(x_i)$ , and then for  $Z = \prod_j Z_j$  in terms of  $y_i f_t(x_i)$ .

**Q 1.6** Show that the 0-1 error is bounded by the exponential cost function  $l(x, y) = e^{(-yf(x))}$ . Deduce that  $Z$  is an upper bound for the 0-1 error.

**Q 1.6.1** Conclude on the exponential decay of the error.

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**Exercice 2 – Entropy**


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**Q 2.1** We roll a biased die  $n$  times independently, where the probability of each side is  $p_k, k = 1..6$ .

**Q 2.1.1** Express the probability of obtaining the sequence  $(x_1, \dots, x_n)$  in terms of the  $p_k$  and  $n_k$  - the number of times that  $k$  appears in the roll.

**Q 2.1.2** What does  $n_k$  tend to as  $n$  approaches infinity? Deduce an expression for the probability of a "typical" sequence in terms of the entropy  $H(p) = -\sum_k p_k \log_2(p_k)$ . What do you notice for high/low values of  $H$ ?

**Q 2.2** Some properties of the entropy function. We call a probability vector a vector representing a discrete probability distribution with  $n$  outcomes :  $\mathbf{p} = (p_1, \dots, p_n)$  such that  $\sum_{i=1}^n p_i = 1$  and  $p_i \geq 0$ .

**Q 2.2.1** Show that  $H(\mathbf{p}) \geq 0$ .

**Q 2.2.2** Let  $\mathbf{p}$  and  $\mathbf{q}$  be two probability vectors of the same dimension.

First, show that  $\log(x) \leq x - 1$  using the fact that the logarithmic function is concave. Deduce that  $-\sum_{i=1}^n p_i \log(p_i) \leq -\sum_{i=1}^n p_i \log(q_i)$ .

**Q 2.2.3** Deduce that  $H(\mathbf{p}) \leq \log(n)$  for a vector  $\mathbf{p}$  of dimension  $n$ .

**Q 2.3** We define the Kullback-Leibler distance as the function  $D(\mathbf{p}||\mathbf{q}) = \sum_{i=1}^n p_i \log(\frac{p_i}{q_i})$

**Q 2.3.1** Show the Jensen inequality : if  $f$  is a convex differentiable function,  $\mathbf{p}$  is a probability vector, and  $t_i$  are any real numbers, then  $\sum_{i=1}^n p_i f(t_i) \geq f(\sum_{i=1}^n p_i t_i)$  (hint : proceed by induction and consider  $p'_i = \frac{p_i}{1-p_n}$ ).

**Q 2.3.2** Deduce that for a convex function,  $f(\mathbb{E}(X)) \leq \mathbb{E}(f(X))$ . Find another proof for question 2.2.2

**Q 2.3.3** Show that  $D(\mathbf{p}||\mathbf{q}) \geq 0$  with equality if and only if  $\mathbf{p} = \mathbf{q}$ . Is  $D$  a distance?

**Q 2.3.4** Consider  $x_1, \dots, x_N$  as i.i.d. observations in  $\mathcal{X}$ , a finite discrete set. The empirical distribution observed is defined by  $\hat{p}(x) = \frac{1}{N} \sum_{i=1}^N \delta(x - x_i)$ , with  $\delta$  being the Dirac function (0 everywhere except at 0 where it is 1). Let  $p_\theta$  be a distribution parameterized by  $\theta$ . Show that maximizing the likelihood is equivalent to minimizing  $D(\hat{p}||p_\theta)$ .